

DeCAL: Towards Physically-Grounded Dexterous Vision-Language-Action Models via Contact-Aware Latent Co-Imagination

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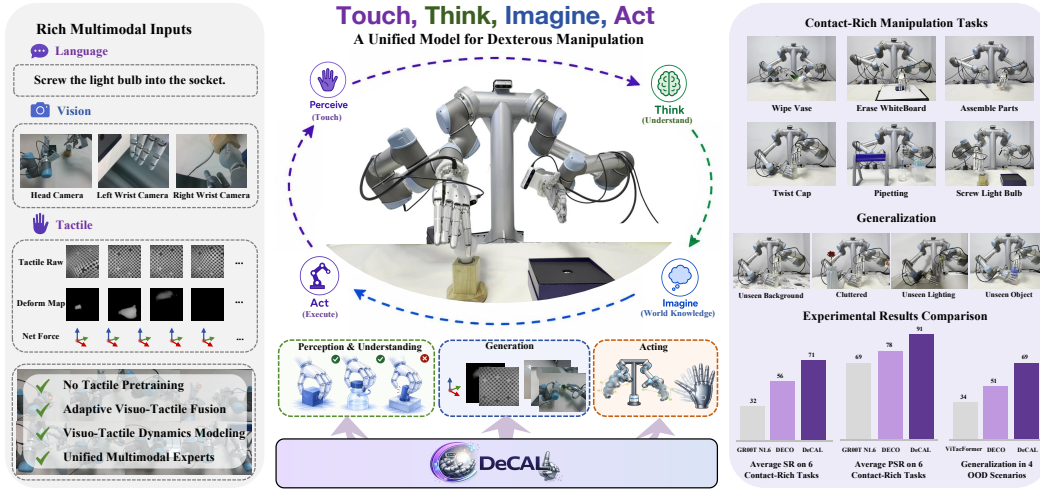


Figure 1: We present DeCAL, a physically-grounded dexterous VLA model that unifies perception, understanding, imagination, and action within a framework. Equipped with rich multimodal inputs, DeCAL achieves strong performance across diverse contact-rich dexterous manipulation tasks and demonstrates robust generalization to unseen scenarios.

Abstract: Dexterous manipulation involves contact-rich and fine-grained interactions with the physical world, posing significant challenges for existing vision-language-action (VLA) models due to severe visual occlusions and complex contact dynamics. While recent works have incorporated tactile sensing into robotic manipulation, most approaches still rely on homogeneous multimodal fusion, lacking adaptive tactile integration and explicit modeling of physical dynamics. In this work, we present DeCAL, a physically-grounded dexterous vision-language-action model that unifies understanding, imagination and action generation for contact-rich dexterous manipulation. Built upon a Mixture-of-Transformers (MoT) architecture, DeCAL leverages specialized experts for each capability while enabling efficient information flow among them. To effectively leverage tactile information, we introduce Adaptive Visuo-Tactile Fusion that dynamically regulates tactile interactions via a contact-aware gating strategy. Furthermore, we propose Visuo-Tactile Latent Co-Imagination to jointly model visual and tactile dynamics, equipping the policy with implicit physical world knowledge. Experimental results show that DeCAL consistently achieves state-of-the-art performance across all tasks, attaining a 71% average success rate and an 83.4% progress success rate, while also demonstrating strong generalization to unseen scenarios.

Keywords: Visuo-Tactile Learning, VLA, Dexterous Manipulation

1 Introduction

Dexterous manipulation plays a fundamental role in human daily life [1, 2], enabling fine-grained and contact-rich interactions with the physical world. Motivated by recent advances in vision-language models [3, 4, 5, 6], vision-language-action (VLA) models have achieved remarkable progress in robotic manipulation [7, 8, 9, 10], exhibiting strong capabilities in visual perception and semantic understanding. Building upon large-scale egocentric human demonstrations [11, 12, 13], recent works [14, 15, 16, 17] have extended VLA models to dexterous manipulation and achieved promising results on multi-fingered manipulation tasks. However, these approaches still struggle with contact-rich and fine-grained interactions due to severe visual occlusions, complex contact dynamics, and limited physical observability from visual inputs alone.

Tactile feedback [18, 19, 20, 21] provides direct access to physical interaction signals, including contact state, force variation, slip, and object deformation, which is essential for dexterous manipulation [22, 23, 24, 25, 26]. Some studies [27, 28, 29] have incorporated tactile sensing into robotic manipulation systems to improve contact awareness and manipulation robustness [30, 31, 32]. However, most existing approaches primarily treat tactile inputs as auxiliary sensory signals and perform homogeneous multimodal fusion, lacking adaptive tactile interaction modeling. Moreover, existing methods are largely reactive and fail to explicitly model future interactions, limiting their ability to reason about evolving contact transitions during manipulation.

In this work, we investigate these problems by introducing DeCAL, a physically-grounded dexterous vision-language-action model that enables effective utilization of tactile signals for contact-rich manipulation. First, DeCAL is built upon a Mixture-of-Transformers architecture composed of three collaborative experts for understanding, imagination, and action generation. The three experts perform directional knowledge sharing through joint attention, providing implicit guidance for coherent and fine-grained dexterous manipulation. Second, we introduce an adaptive visuo-tactile fusion mechanism that dynamically injects tactile information via a contact-aware gating strategy, allowing the model to selectively emphasize tactile cues during physical interactions. Finally, we propose visuo-tactile latent co-imagination, which jointly models visuo-tactile dynamics through future-oriented latent imagination, equipping the policy with implicit world knowledge.

To comprehensively evaluate DeCAL, we conduct extensive real-world experiments on diverse contact-rich dexterous manipulation tasks. The comparative results demonstrate that DeCAL consistently achieves state-of-the-art performance, outperforming the strongest baseline by more than 15% average success rate. Meanwhile, DeCAL maintains efficient real-time performance, achieving an average inference latency of 0.27s per action chunk. Furthermore, comprehensive ablation studies verify the effectiveness of each proposed component, highlighting the promise of effectively leveraging tactile information for dexterous manipulation. In summary, our contributions are as follows:

- We present DeCAL, a dexterous Vision-Tactile-Language-Action framework that unifies understanding, generation, and action for effective visuo-tactile representation learning.
- We introduce Adaptive Visuo-Tactile Fusion and Visuo-Tactile Latent Co-Imagination to facilitate contact-rich and fine-grained dexterous manipulation.
- We demonstrate the effectiveness and generalization of our method through a range of real-world experiments.

2 Related Work

2.1 Vision-Language-Action Model

Vision-Language-Action (VLA) models have emerged as a powerful paradigm for robotic control [7, 33, 9, 17, 15, 34], leveraging the broad semantic and world knowledge embedded in large-scale pre-trained foundation models [4, 5, 6]. By fine-tuning these models on diverse robotic datasets [35, 36, 37, 38], researchers have demonstrated successful cross-domain capability transfer from general

intelligence to specific manipulation tasks. Representative examples include $\pi_{0.5}$ [7], and GR00T N1.6 [39]. More recently, a growing line of work, including InternVLA-A1 [40], MoTus [41], and LAST₀ [42], has advanced VLA research by adopting Mixture-of-Transformer (MoT) architectures. These approaches seamlessly integrate perception (understanding), generation, and action within a unified framework, achieving strong performance across a variety of complex manipulation tasks. Despite these advances, existing VLA models remain predominantly vision-centric and lack explicit modeling of physical contact dynamics, limiting their performance in tactile-intensive scenarios involving severe visual occlusions and fine-grained physical interactions. To address this, we integrate tactile sensing into VLA world modeling by jointly learning visual evolution and contact dynamics, enabling more fine-grained and physically grounded manipulation.

2.2 Tactile for Dexterous Manipulation

Dexterous manipulation poses a fundamental challenge for vision-only policies, as the intricate physical structure of multi-fingered hands often leads to severe self-occlusion and ambiguous contact observations [2, 43]. To mitigate this, recent studies have explored the integration of haptic feedback to maintain operational continuity under visual impairment [44, 45, 46, 47, 22, 23]. For instance, some studies [29, 48, 49, 50] learn rich tactile representations through self-supervised objectives, capturing contact geometry and force-related cues to facilitate efficient downstream policy learning. Another line of work [27, 28] predicts future tactile signals to guide action generation, showing the potential of tactile foresight for improving visuo-tactile coordination and contact-aware control. Despite these advances, how to effectively and adaptively couple visual and tactile signals still remains underexplored [45, 31]. Distinct from these approaches, our work integrates tactile sensing into VLA policy learning, leveraging the model’s multimodal understanding of language, vision, and physical interactions [51]. By introducing a unified dynamics modeling framework and an adaptive learning mechanism, our model enables more robust visuo-tactile fusion, allowing the agent to dynamically prioritize visual and tactile cues according to task demands and environmental constraints.

3 Robot System Setup

Our system consists of a pair of 6-DoF UR5 robotic arms and two 22-DoF SharpaWave five-fingered dexterous hands. Visual observations are captured from two wrist-mounted cameras and one egocentric camera, all using Intel RealSense D435. Each fingertip is equipped with a high-resolution (320×240) vision-based tactile sensor developed by Sharpa, enabling fine-grained perception of contact dynamics during manipulation. Specifically, a built-in camera inside each sensor captures the deformation of the elastic surface during physical interaction. Based on the raw tactile images and visuo-tactile processing algorithms, we represent tactile signals in three complementary forms:

- **Raw Image (R).** Raw tactile observations are directly captured by the built-in camera inside each visuo-tactile sensor, recording the contact patterns and elastic surface deformations.
- **Net Force (F).** Each fingertip outputs a 6-DoF force representation consisting of 3-axis forces and torques. The force signals are estimated from raw tactile observations using a pretrained regression model, enabling force-aware tactile perception.
- **Deform Map (M).** Each fingertip outputs a deformation depth map represented as a 2D image through a translation model [52], where each pixel indicates the local deformation depth on the tactile surface, providing fine-grained geometric contact information.

4 Method

4.1 Preliminaries

The goal of our robot policy is to predict an action chunk from multimodal observations. Most existing VLA models lack explicit modeling of future interaction dynamics, limiting their ability to

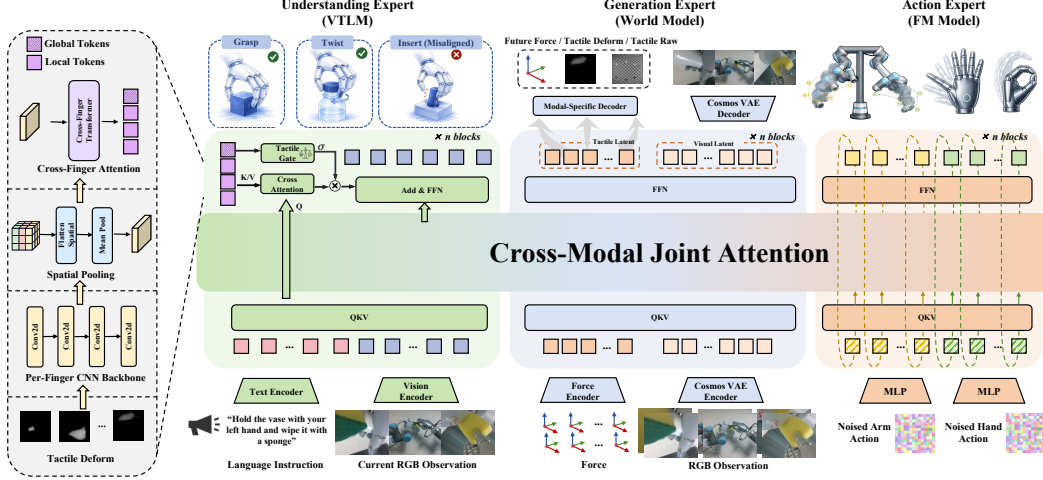


Figure 2: DeCAL is built upon a MoT architecture that unifies scene understanding, visuo-tactile dynamics foresight, and action generation. The Action Expert employs Factorized Flow Matching to decouple arm and hand motion, enabling better coordination and dexterous manipulation.

reason about fine-grained physical interactions. To address this, we propose DeCAL π_θ , a unified VLA framework that jointly performs understanding, imagination, and action generation. Given the current observation $o_t = (I_t, H_t, s_t)$, and the language instruction l , where I_t denotes the visual observations, H_t denotes the tactile observations, and s_t represents the robot state. DeCAL is trained to jointly maximize the likelihood of future visuo-tactile latent z_{t+H} and future actions $a_{t+1:t+H}$:

$$\max_{\theta} \mathbb{E}_{(o_t, l, a_{t+1:t+H}, z_{t+H}) \sim \mathcal{D}} [\log \pi_{\theta}(a_{t+1:t+H}, z_{t+H} \mid o_t, l)]. \quad (1)$$

4.2 Model Architecture

DeCAL adopts a Mixture-of-Transformers (MOT) architecture that seamlessly integrates scene understanding, visuo-tactile dynamics foresight, and action generation within a unified framework. As illustrated in Figure 2, the framework consists of three specialized transformer experts.

Understanding Expert. The understanding expert is built upon Qwen3-VL [3] for its strong multimodal understanding capability. Given language instructions and multi-view visual observations, the inputs are first encoded into text and visual tokens, which are then processed by transformer blocks to produce contextual embeddings shared with downstream experts through masked self-attention. We further extend the multimodal inputs to tactile observations through a dedicated cross-attention mechanism, where tactile features serve as keys and values, and visual-language features act as queries. The resulting representations are then residually fused with the self-attention features, enabling joint reasoning over visual, linguistic, and tactile observations within a unified latent space.

Generation Expert. The Generation Expert explicitly models visuo-tactile dynamics by predicting future visual and tactile observations from historical interactions, forming physically grounded multimodal world representations to guide downstream action generation. Despite the remarkable progress of recent video generation models in visual prediction, directly applying generative modeling to contact-rich manipulation remains challenging due to the strict real-time requirements of high-frequency robot control. Following Cai et al. [40], we adopt a parallel decoding strategy for future visuo-tactile generation. The proposed non-autoregressive paradigm significantly improves computational efficiency while remaining effective for modeling future visuo-tactile interactions.

Action Expert. Conditioned on the latent features from the understanding expert and generation expert, together with the robot proprioceptive states q_t , the action expert predicts an action chunk $a_{t:t+H}$ using a flow matching objective. Since arm and hand motions exhibit substantially distinct dynamics and control granularity, jointly modeling them through a shared denoising process may obscure fine-grained dexterous behaviors and weaken arm-hand coordination in high-DoF settings.

To address this, we propose **Factorized Flow Matching**, which explicitly decouples arm and hand motion generation. Specifically, arm and hand actions are initialized from independent noise distributions and projected into separate token sequences through MLP layers. The tokens are jointly processed by transformer blocks for coordinated motion generation before being decoded into clean action trajectories. Both the generation expert and action expert adopt Qwen3 [53] as their backbone.

Cross-Modal Joint Attention. We implement a blockwise attention mask to control the information flow across experts, as shown in Figure 3. Information is propagated unidirectionally from the understanding expert to the generation expert, and then to the action expert. Tokens in later experts can attend to tokens from preceding experts, while the reverse attention is prohibited. Within both the understanding expert and generation expert, tokens are bidirectionally attended for more effective multimodal reasoning and representation learning. In the action expert, state tokens can attend to themselves and all earlier blocks, while action tokens can attend to themselves, state tokens, and all preceding blocks. This asymmetric attention design enables action generation to fully condition on semantic understanding, imagined future interactions, and robot states.

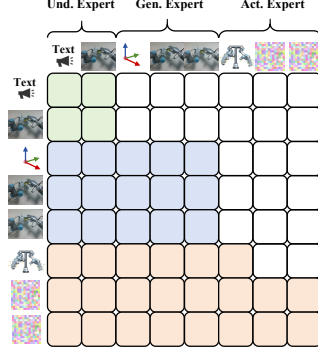


Figure 3: **Attention mask.**

4.3 Adaptive Visuo-Tactile Fusion

Tactile Feature Extraction. We use multi-finger deform maps as tactile inputs due to their rich geometric contact information. A shared ResNet-based [54] tactile encoder extracts tactile features from all fingertips. The extracted features are spatially pooled and projected into tactile tokens, which are subsequently fused through transformer layers to capture cross-finger interactions and global contact patterns. The tactile encoder finally produces two complementary representations: local tactile tokens that preserve fine-grained contact details for visuo-tactile interaction, and a global tactile token that summarizes the overall contact state for contact-aware tactile gating.

Adaptive Visuo-Tactile Cross-Attention. To incorporate tactile feedback into visual-language embeddings, we introduce a cross-attention mechanism at each transformer block. The local tactile tokens serve as keys and values, while the prefix projection of the transformer input provides the queries. The resulting fused output is then added to the embeddings from the block’s joint self-attention via a residual connection, allowing the model to integrate tactile cues while preserving the original multimodal context.

Contact-Aware Gating Mechanism. In dexterous manipulation, physical contact is often intermittent and only arises during specific phases of a task, rather than being continuously present. Directly injecting tactile signals into the policy at all times can introduce biases in action generation and perturb the distribution of visual attention. To address this, we introduce a Contact-Aware Gating Mechanism that allows the policy to adaptively modulate the weights of tactile inputs based on the relevance of contact information at each timestep. Formally, the gated output is computed as:

$$\begin{aligned}\tilde{X} &= X + \sigma \odot \text{CrossAttn}(Q_{vl}, K_{local}, V_{local}), \\ \sigma &= \text{Gate}(z_{global}),\end{aligned}\tag{2}$$

where X represents the original input embeddings to the transformer layer, Q_{vl} is the query from visual-language embeddings, while K_{local} and V_{local} correspond to the local tactile tokens serving as keys and values. The gating weight σ is derived from the global tactile token z_{global} through a lightweight projection function $\text{Gate}(\cdot)$. This mechanism allows the model to selectively incorporate tactile feedback based on the overall contact state across all fingertips.

4.4 Visuo-Tactile Latent Co-Imagination

The generation expert performs future-oriented latent imagination to jointly model visuo-tactile dynamics. Given historical and current visuo-tactile observations, the model produces predictive latent representations that capture the underlying physical dynamics of the interaction.

Tactile Latent Prediction. 6-DoF contact force provides a compact and physically meaningful representation of contact dynamics, capturing both force magnitude and torque information. For tactile latent prediction, the forces from each fingertip are first mapped into embeddings using a dedicated force encoder, which are then processed by the policy to produce a tactile latent. The resulting latent can subsequently be decoded into representations of different granularities, including raw images, deform maps, and 6-DoF force vectors. These representations capture tactile dynamics from multiple perspectives, supporting physically grounded scene understanding and action generation.

Visual Latent Prediction. Inspired by recent world action models [55, 56, 57], we equip DeCAL with world knowledge by generating future visual latents from historical and current observations. This forward-looking representation provides temporally informed context that enhances planning and decision-making in contact-rich manipulation tasks. We encode the multi-view visual images using the CosMOS VAE tokenizer [58], which provides expressive visual latents while preserving fine-grained details. To efficiently handle long visual sequences, we further apply a token compression mechanism following Cai et al. [40]. Specifically, the 32×32 latent feature grid is downsampled to 4×4 using a convolutional layer with an 8×8 kernel, reducing sequence length while retaining essential spatial information. The compressed tokens are then decoded in parallel to generate future visual frames, leveraging the KV cache from the understanding expert.

5 Experiment

In this section, we evaluate DeCAL through extensive experiments addressing two key questions: (1) How does DeCAL perform against state-of-the-art policies and generalize to OOD scenarios (Section 5.2 and 5.3)? (2) How does each component contribute to the overall performance (Section 5.4)?

5.1 Experiment Setup

Tasks. We evaluate DeCAL on six contact-rich and fine-grained dexterous manipulation tasks, as illustrated in Figure 4: (1) *Wipe Vase*, (2) *Erase Whiteboard*, (3) *Assemble Parts*, (4) *Twist Cap*, (5) *Pipetting*, (6) *Screw Light Bulb*. Each task is collected with 100 high-quality demonstrations and evaluated with 20 trials by default. We collect expert demonstrations through a teleoperation system with MetaGlove Pro and VIVE Trackers, following Fu et al. [17], where the gloves retarget human hand motion to dexterous hands and the trackers control the robotic arm end-effectors. More details about the teleoperation setup and task specifications are provided in the Appendix.

Baselines and Evaluation Metrics. We compare DeCAL with two state-of-the-art VLA models, GR00T N1.6 [39] and InternVLA-A1 [40], as well as two tactile-based specialist policies, ViTacFormer [27] and DECO [31], which are built upon different architectures. To enable a fairer comparison in terms of tactile modality usage, we further reproduce InternVLA-A1^t by naively incorporating tactile deform maps and net-force signals into the VLM backbone. We use two metrics to evaluate model performance: Success Rate (SR), indicating the entire task is successfully completed, and Progress Rate (PSR), capturing the average completion ratio across all task stages.

5.2 Results and Analysis

Results on Real-World Experiments. As shown in Table 1, DeCAL achieves the highest success rate across all tasks, outperforming all baselines, while also attaining the best PSR on the majority of tasks. Vision-based policies perform reasonably well on simpler tasks, such as *Wipe Vase* and *Pipetting*. However, they struggle with contact-rich dexterous manipulation requiring precise physical interaction and fine-grained contact understanding. Policies equipped with tactile sensing can incorporate richer environmental feedback, improving physical interaction awareness. But naively injecting tactile signals may also disturb the distribution of visual representations, leading to reduced accuracy in visual grounding and grasping. This issue becomes particularly evident in the *Assemble Parts* task, where effective coordination between vision and tactile is crucial. Benefiting from the adaptive visuo-tactile fusion mechanism and the joint modeling of visuo-tactile dynamics, DeCAL enhances

Table 1: **Main results of six real-world tasks.** Each experiment is evaluated with 20 trials.

Method	Wipe Vase		Erase Whiteboard		Assemble Parts		Twist Cap		Pipetting		Screw Light Bulb	
	SR	PSR	SR	PSR	SR	PSR	SR	PSR	SR	PSR	SR	PSR
GR00T N1.6	30.0%	78.8%	50.0%	55.0%	30.0%	60.0%	10.0%	46.7%	60.0%	85.0%	10.0%	45.0%
InternVLA-A1	65.0%	85.0%	50.0%	58.3%	15.0%	43.3%	5.0%	36.7%	35.0%	78.8%	30.0%	53.8%
ViTacFormer	80.0%	88.8%	45.0%	50.0%	15.0%	71.7%	65.0%	81.7%	55.0%	86.3%	25.0%	46.3%
DECO	90.0%	95.0%	60.0%	70.0%	35.0%	61.7%	70.0%	83.3%	45.0%	71.3%	35.0%	41.3%
InternVLA-A1 [†]	75.0%	87.5%	45.0%	51.7%	45.0%	65.0%	25.0%	55.0%	20.0%	58.8%	25.0%	31.3%
DeCAL (Ours)	100.0%	100.0%	80.0%	86.7%	65.0%	85.0%	80.0%	93.3%	60.0%	86.3%	40.0%	48.8%

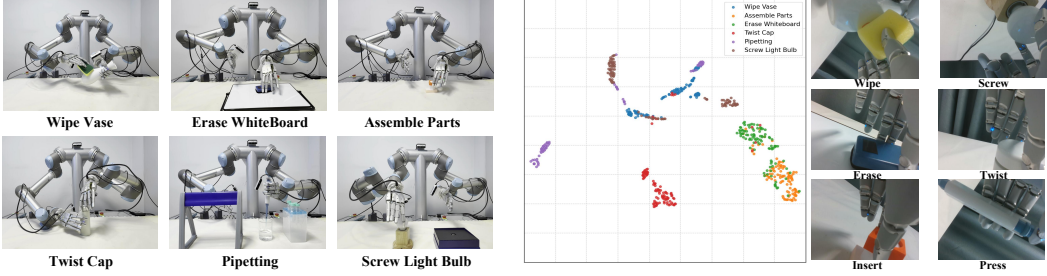


Figure 4: **Visualization of dexterous tasks.** Figure 5: **Visualize t-SNE of global tactile tokens.**

physical interaction awareness while preserving robust visual grounding capabilities, thereby leading to improved performance on fine-grained dexterous manipulation tasks.

Tactile Latent Analysis. We analyze the global tokens extracted by the tactile encoder using t-SNE [59]. For each task, we sample frames from different contact phases, which correspond to diverse interaction patterns such as twisting, insertion, and wiping. As illustrated in Figure 5, the learned global tactile representations exhibit clear clustering behavior under different contact modes, indicating that the encoder is able to capture meaningful and structured physical interaction patterns.

Adaptive Tactile Gate. Next, we evaluate the effectiveness of the adaptive tactile gating mechanism. Taking the Assemble Parts and Twist Cap tasks as examples, Figure 6 visualizes the tactile gate outputs across different frames within a testing episode. When the hand is not interacting with the environment, the tactile gate values remain consistently low, indicating that the policy is primarily dominated by visual signals. In contrast, once meaningful physical contact occurs between the fingers and the objects, the gate values increase significantly, allowing tactile information to play a more prominent role in the policy and influence action generation.

Tactile Force Prediction. We further compare the predicted tactile forces with the ground-truth values in the validation data. As shown in Figure 7, the predicted 6D forces highly align with the ground-truth variations across different contact stages. This demonstrates that the proposed visuo-tactile latent co-imagination can effectively model underlying contact dynamics from multimodal observations. Moreover, the learned dynamics are further propagated to the action expert through joint attention, enabling more stable and precise action generation for contact-rich dexterous manipulation.

5.3 Generalization

Besides the remarkable effectiveness, DeCAL also demonstrates promising generalization capabilities under four *out-of-distribution* scenarios: (1) **Unseen Background**, where a tablecloth perturbs the visual distribution. (2) **Cluttered Environment**, where random objects introduce visual distractions. (3) **Unseen Lighting**, where we use dimmer lighting conditions to simulate different illumination

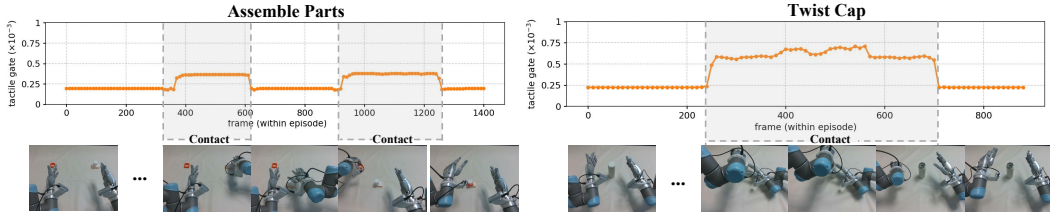


Figure 6: **Tactile gate.** The tactile gate value σ increases during contact phases.

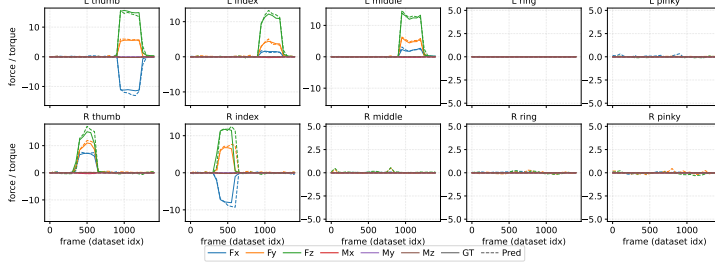


Figure 7: Force prediction.

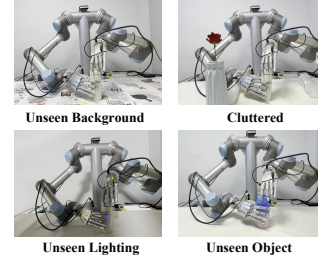


Figure 8: OOD scenarios.

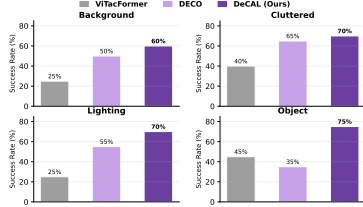


Figure 9: Generalization results.

Table 2: Ablation study of each components.

Factorized FM	Tac Gating	Vis Gen	Tac Gen	Assemble Parts	Twist Cap
✗	✓	✓	✓	25.0%	35.0%
✓	✓	✗	✗	20.0%	30.0%
✓	✓	✗	✓	35.0%	45.0%
✓	✓	✓	✗	55.0%	60.0%
✓	✗	✓	✓	50.0%	70.0%
✓	✓	✓	✓	65.0%	80.0%

settings. (4) **Unseen Object**, where a novel cup with different shape, diameter, and height is introduced for manipulation, as shown in Figure 8. Taking Twist Cap as an example, we report the success rate of DeCAL, DECO, and ViTacFormer under four OOD scenarios in Figure 9. The results demonstrate that DeCAL generalizes effectively across various distribution shifts, achieving a 75.0% success rate under the unseen object setting and substantially outperforming all baselines.

5.4 Ablation Studies

In this section, we analyze the contribution of each component in DeCAL to the overall policy performance. We conduct ablation studies focusing on four aspects: (1) Factorized Flow Matching; (2) Tactile Gating Mechanism; (3) Visual Latent Generation; (4) Tactile Latent Generation. As shown in Table 2, Factorized Flow Matching contributes significantly to the overall task performance. Without this mechanism, we observe noticeable inconsistency and poor coordination between the arms and dexterous hands during manipulation. In addition, visual and tactile latent generation effectively models motion and contact dynamics, leading to more accurate action generation. Finally, the adaptive tactile gating mechanism enables the policy to dynamically adjust the importance of visual and tactile cues across different interaction stages, further improving the success rate on contact-rich manipulation tasks. More quantitative and qualitative results are provided in the Appendix.

6 Conclusions

In this paper, we present DeCAL, a physically-grounded dexterous vision-language-action model that effectively leverages tactile feedback for contact-rich manipulation. By unifying understanding, generation, and action within a collaborative framework, DeCAL enables coherent multimodal reasoning and fine-grained dexterous control. Furthermore, DeCAL adaptively fuses visuo-tactile information through contact-aware tactile gating and models visuo-tactile dynamics via future-oriented latent imagination, endowing the policy with implicit world knowledge. Extensive experiments on real-world dexterous tasks demonstrate the effectiveness and generalization of our approach.

7 Limitations

Our method relies on accurate tactile perception, which can be affected by sensor noise, calibration errors, and model drift, potentially leading to performance degradation during long-term and high-load operation. Moreover, our current framework does not leverage large-scale visuo-tactile pretraining. We believe that scaling up diverse tactile dexterous data for pretraining could further improve the model’s capability, which we consider an important direction for future research.

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